Robust efficiency of the Smears–Vohralík-type estimator for algebraically stabilised convection–diffusion–reaction

galerkit, gap 1 of litscan_2026-10.md. Estimator: `sv_estimator` (sv_convection_extension.md). 18 October 2026.

0. Summary and status

Results.

- **Theorem 1 (reliability).** The estimator is a guaranteed upper bound for E = |||e|||_⊕ + ‖S˜‖_*, for every σ ≥ 0 and every conforming u_h. The proof is complete and was given before; it is recalled here.
- **Theorem 2 (efficiency).** For each of the two functionals the estimator equilibrates, ℓ = Res_A and ℓ = −S˜:
  - the local contribution is bounded by the local dual norm of ℓ on a vertex neighbourhood, plus data oscillation;
  - the global estimator is bounded by ‖ℓ‖_* plus oscillation.
  - The constants depend only on d and the shape regularity. They are independent of ε, σ, b, c, h, the stabilisation tensor A, and u_h.
- **Corollary 3.** The estimator is two-sided and robust in E: E ≤ η_E ≤ C(1 + c_b)²(E + osc). In the weighted measure E_c of Corollary 3' the factor is linear, C(1 + c_b). (Corrected, audit G2: the earlier (1 + c_b) was wrong for c_b > 1.)

What the proof relies on. It is complete given four cited results:

- the inverse inequalities and patch stability of [SV20, Lemmas 2.2 and 4.1];
- Verfürth's squeezed-bubble estimates [V05, Lemma 3.6];
- the stability of a patch-mean quasi-interpolant [V05, Lemma 3.3];
- an L²-bounded commuting projection onto RT₀ with mixed boundary conditions [CW08; Licht 2019]. Licht's abstract confirms commuting smoothed projections that preserve homogeneous boundary conditions on a boundary part and are uniformly bounded; the L² form of the bound is to be confirmed in the full text (audit T3).

Only the last one is outside the papers already read. It is used in Lemma H(ii) and Proposition 5.2 (Remark 5.4), for σ > 0 and σ = 0 alike. For Γ_N = ∅ the projection of [CW08] suffices; mixed boundary conditions need [Licht 2019] (corrected, audit T6).

Departures from the published arguments. The new ingredients are marked ◆ in the text:

1. **Weight-free construction (Lemma G).** The patch problems carry no SV weights; the weights enter only through the evaluation (Lemma A). The reaction-dominated case needs a new comparison argument: the best-approximation trick (Lemma G, case 1). This is consistent with SV20's statement that the weights are necessary; see Remark A.1.
2. **Stabilisation as a diffusion tensor.** The stabilisation enters through D = εI + A. Compatibility is restored by an explicit defect shift, and its cost is bounded through a quasi-interpolant (Lemma H). This is what lets the bound cover inexact iterates.
3. **Augmented-norm constant (Remark 1.2).** The factor 3 in |||e|||_⊕ ≤ 3‖Res‖_* [EM16, Thm. 4.1] becomes 2 + max{1, c_b} for galerkit's norm. This was found while writing the proof, and the code is corrected (`kappa_aug`).

Not refereed. The result is checked numerically in §6.

1. Setting

Hypotheses

- **(H1)** Ω ⊂ $\mathbb{R}^d$ (d = 2, 3) is a polytope, and T_h is a conforming simplicial mesh with shape-regularity parameter ϑ.
- **(H2)** The coefficients satisfy:
  - ε > 0 is constant;
  - b ∈ W^{1,∞}(Ω)^d, c ∈ L^∞(Ω), f ∈ L²(Ω), g_N ∈ L²(Γ_N).

- **(H3)** $c - \tfrac{1}{2}\operatorname{div} b \geq \sigma \geq 0$, and $c_b := \min\{\|c - \tfrac{1}{2}\operatorname{div} b\|_\infty / \sigma, \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty C_F^2 / \varepsilon\} < \infty$, with C_F the Friedrichs constant of H^1_D (`friedrichs_constant`; the first entry is ∞ if $\sigma = 0$). (Restated, audit N4: the earlier form $\|c - \tfrac{1}{2}\operatorname{div} b\|_\infty \leq c_b\,\sigma$ excluded $\sigma = 0$ with $c \neq \tfrac{1}{2}\operatorname{div} b$, which the code covers by the second entry.) Theorem 2 does not use (H3); it enters only through Remark 1.2.
- **(H4)** The Dirichlet part Γ_D has positive measure and contains the inflow boundary, so $b\cdot n \geq 0$ on Γ_N.
- **(H5)** u_D is the trace of a P1 function. Non-P1 data add the lifting terms of gap 4; their efficiency is not treated here.
- **(H6)** $u_h \in V_h = P1 \cap H^1(\Omega)$ with $u_h = u_D$ on Γ_D. The stabilisation is $S(u_h; u_h, v_h) = (A\nabla u_h, \nabla v_h)$ for all $v_h \in V_{h,0}$, with A cellwise constant, symmetric and positive semidefinite. This holds exactly for every algebraic stabilisation in galerkit (gap 2).
  - **Nothing is assumed about how u_h was obtained.** It may be any iterate.

Norms and functionals

Norms and weights.

- $|||v|||^2_\omega = \varepsilon\|\nabla v\|^2_\omega + \sigma\|v\|^2_\omega$.
- $\|\ell\|_{,\omega} = \sup\{\ell(v) / |||v|||_\omega : v \in H^1_{}(\omega) \setminus \{0\}\}$, with $H^1_*(\omega) = \{v \in H^1(\omega) : v = 0 \text{ on } \partial\omega \setminus \Gamma_N\}$, and $\|\ell\|_* = \sup\{\ell(v) / |||v||| : v \in H^1_D(\Omega) \setminus \{0\}\}$.
  - (Corrected, audit G1: the earlier $H^1_0(\omega)$ is too small. Lemma H(i) tests with ψ_a at compatible Neumann vertices, and Lemma D with face bubbles on Neumann faces; neither vanishes on Γ_N. Zero extension maps $H^1_{}(\omega)$ into $H^1_D(\Omega)$, so A6 and the monotonicity in ω still hold. This is the space of §9, and `localized_error` already uses it.)*
- $\alpha_S = \min\{h_S\,\varepsilon^{-1/2}, \sigma^{-1/2}\}$ for an element or face S, with $\alpha_S = h_S\,\varepsilon^{-1/2}$ if $\sigma = 0$.
- $\theta_S = \varepsilon^{1/2}\,\sigma^{-1/2} / h_S$, so $\alpha_S = h_S\,\varepsilon^{-1/2}\min\{1, \theta_S\}$.

The residual functionals. For $v \in H^1_D$, with $e = u - u_h$:

- $\mathrm{Res}(v) = (f, v) + (g_N, v)_{\Gamma_N} - B(u_h, v) = B(e, v)$;
- $S^{\sim}(v) = (A\nabla u_h, \nabla v)$;
- $\mathrm{Res}_A = \mathrm{Res} - S^{\sim}$, so $\mathrm{Res}_A(v) = (r, v) - ((\varepsilon I + A)\nabla u_h, \nabla v) + (g_N, v)_{\Gamma_N}$, where $r = f - b\cdot\nabla u_h - c\,u_h$.

Both Res_A and $-S^{\sim}$ are of the generic form

$\ell(v) = (r_\ell, v) - (D\nabla u_h, \nabla v) + (g_\ell, v)_{\Gamma_N}$,

with $(r_\ell, D, g_\ell) = (r, \varepsilon I + A, g_N)$ for Res_A and $(0, A, 0)$ for $-S^{\sim}$.

Augmented norm. Write $B = B_S' + B_A'$ with

- $B_A'(w, v) = (b\cdot\nabla w + \tfrac{1}{2}(\operatorname{div} b)\,w, v)$;
- $B_S'(w, v) = \varepsilon(\nabla w, \nabla v) + ((c - \tfrac{1}{2}\operatorname{div} b)\,w, v)$.

Then $|||e|||_\oplus = |||e||| + \|B_A'(e, \cdot)\|_*$, and $E = |||e|||_\oplus + \|S^{\sim}\|_*$.

Remark 1.2 (◆ the factor in the augmented bound).

- By (H3), $|B_S'(w, v)| \leq \varepsilon^{1/2}\|\nabla w\|\,\varepsilon^{1/2}\|\nabla v\| + \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty\|w\|\|v\| \leq \max\{1, c_b\}\,|||w|||\,|||v|||$. Here $\|w\|\|v\| \leq \sigma^{-1}|||w|||\,|||v|||$ for the first entry of c_b, and $\|w\|\|v\| \leq C_F^2\,\varepsilon^{-1}|||w|||\,|||v|||$ (Friedrichs) for the second.
- Coercivity (H3)–(H4) gives $|||e||| \leq \|\mathrm{Res}\|_*$, and $\|B_A'(e, \cdot)\|_* \leq \|\mathrm{Res}\|_* + \max\{1, c_b\}|||e|||$.
- Hence $|||e|||_\oplus \leq \kappa\|\mathrm{Res}\|_*$, with $\kappa = 2 + \max\{1, c_b\}$.
- EM16's $\kappa = 3$ is the case $c - \tfrac{1}{2}\operatorname{div} b \equiv \sigma$. All constant-coefficient examples in galerkit have $c_b = 1$. AJ27 Ex. 4.3 (c variable, $\sigma = \min c \approx 3\cdot 10^{-21}$) does not, and its former augmented bound was not guaranteed.
- `abr17_estimator` and `sv_estimator` now use κ (`kappa_aug`). When σ does not control $c - \tfrac{1}{2}\operatorname{div} b$, they take the Friedrichs alternative $c_b = \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty C_F^2/\varepsilon_{\min}$, but only if it lowers $\max\{1, c_b\}$. C_F is the Friedrichs constant of H^1_D, from `friedrichs_constant`:
  - **no Neumann boundary:** the H^1_0 constant of the bounding box, valid on any domain by zero extension;

- **Neumann boundary on a box domain:** $C_F^{-2} = \Sigma_i \lambda_i$, with $\lambda_i = (\pi/L_i)^2$ if both faces normal to e_i are entirely Dirichlet, $(\pi/2L_i)^2$ if one is, and 0 otherwise. This follows by separation of variables, and counting a partly Neumann face as Neumann only lowers λ. For AJ27 Ex. 4.3 (Neumann at $x = 1$) it gives $C_F^2 = 1/(1.25\pi^2)$, and $\kappa \approx 8.1 \cdot 10^4$ is now guaranteed; the earlier $5 \cdot 10^4$ used the H^1_0 constant and was not;
- **Neumann boundary on another domain:** no guaranteed constant is computed. For $\sigma > 0$ the σ-based c_b is kept. For $\sigma = 0$ the bounding-box constant is used and flagged as not guaranteed, also when it brings $\max\{1, c_b\}$ down to 1.

The same C_F enters η_{alg} and the Dirichlet-lifting terms.

- Conversely, $\|Res\|_* \le \max\{1, c_b\}|||e||| + \|B_A'(e, \cdot)\|_* \le \max\{1, c_b\}|||e|||_\oplus$.

The estimator (as implemented)

For each vertex a, with patch ω_a and hat function ψ_a, define:

- V^a: RTN_1 on T_a, $H(div)$-conforming. Its normal component is:
  - 0 on $\partial\omega_a \setminus \Gamma_D$ (on all of $\partial\omega_a$ if a is interior);
  - $-\Pi_F(\psi_a\, g_\ell)$ on Neumann faces;
  - free on Dirichlet faces.
- $g_a = \Pi_1(\psi_a\, r_\ell - \nabla\psi_a \cdot D\nabla u_h)$.
- The defect $\delta_a = \ell(\psi_a)$ if T_a has no free face ("compatible" patch), and $\delta_a = 0$ otherwise. Set $c_a = \delta_a / |\omega_a|$ and $\tilde{g}_a = g_a - c_a$.
- **Quadrature defect** (audit R1). For $\ell = Res_A$, δ_a is computed with the estimator's quadrature (degree 6), while the discrete system is assembled with its own rule (degree 4 by default). Write $\delta_a = \delta_a^{alg} + \delta_a^{quad}$:
  - $\delta_a^{alg} = -r_a$, with r the assembled algebraic residual. It vanishes at the discrete solution.
  - $\delta_a^{quad} = Res_A^{est}(\psi_a) - Res_A^{asm}(\psi_a)$ is the difference of the two rules on all data terms tested with ψ_a: f, $b \cdot \nabla u_h$, $c\, u_h$, the diffusion (with $\Pi_1\varepsilon$ in §11) and g_N. The stabilisation term is exact.
  - $\delta_a^{quad} = 0$ when the assembly rule integrates every data term exactly. With the degree-4 rule and P1 tests, that means ε of degree ≤ 4, f and b of degree ≤ 3, c of degree ≤ 2, and g_N integrated exactly by the assembly's face rule. Otherwise it is a data-quadrature oscillation, of the order of the quadrature error.
  - Its spread $\psi^{quad} = \Sigma_a \delta_a^{quad} \chi_{\omega_a}/|\omega_a|$ has the dual norm $osc_q := \|\psi^{quad}\|_*$.
  - Smoke check (constant ε, smooth data, converged solution): $\eta_{def,A} = 1.5 \cdot 10^{-7}$, against $2.4 \cdot 10^{-11}$ from δ^{alg} alone, with $\eta_A = 0.63$.
  - Reliability is unaffected, because the full δ_a is shifted and bounded.

Patch problems.

- $\sigma > 0$: $\sigma^a = \operatorname{argmin}$ over $v \in V^a$ of $J_a(v) = \varepsilon^{-1}\|\psi_a D\nabla u_h + v\|^2_{\omega_a} + \sigma^{-1}\|\operatorname{div} v - \tilde{g}_a\|^2_{\omega_a}$.
- $\sigma = 0$: the same flux term is minimised subject to $\operatorname{div} v = \tilde{g}_a$. This is solvable, because the shift makes $(\tilde{g}_a, 1)_{\omega_a}$ consistent with the prescribed normal fluxes.

Notation. $\tau^a = \psi_a D\nabla u_h + \sigma^a$ and $m^a = \operatorname{div}\sigma^a - \tilde{g}_a$, with $m^a = 0$ if $\sigma = 0$. Then $\sigma_h = \Sigma_a \sigma^a$ and $\tau = D\nabla u_h + \sigma_h = \Sigma_a \tau^a$ on each K.

Estimator.

- On each cell: $\eta_{\ell,K} = \min\{\varepsilon^{-1/2}\|\tau\|_K, \Sigma_{F\subset\partial K} c_F\|\tau\cdot n\|_F + m_K\|\operatorname{div}\tau\|_K\} + \sigma^{-1/2}\|\Sigma_a m^a\|_K + m_K\|r_\ell - \Pi_1 r_\ell\|_K$, where
  - $m_K = \min\{h_K/(\pi\,\varepsilon^{1/2}), \sigma^{-1/2}\}$;
  - $c_F^2 = (|F|/|K|)\min\{h_K^2(1/\pi^2 + 2/(d\pi))/\varepsilon,\ 1/\sigma + h_K/(d(\sigma\varepsilon)^{1/2})\}$.
- Total: $\eta_\ell = (\Sigma_K \eta^2_{\ell,K})^{1/2} + \eta_{\ell,def}$.
- The defect term is $\eta_{\ell,def} = \min$ over $s \in RT_0$ with $s\cdot n = 0$ on Γ_N of $(\varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|\psi_{def} - \operatorname{div} s\|^2)^{1/2}$, with $\psi_{def} = \Sigma_a c_a \chi_{\omega_a}$. For $\sigma = 0$ the constraint $\operatorname{div} s = \psi_{def}$ is imposed instead.

- Then η_A = η_{Res_A}, η_S = min{gap 2/6 bound, η_{−S˜}}, and η_E = κ(η_A + η_S) + η_S, plus the boundary terms.

2. Results

Theorem 1 (reliability). Under (H1)–(H6):

- $\|\ell\|_* \le \eta_\ell + \eta_N(\ell)$ for $\ell \in \{\mathrm{Res}_A, -S\tilde{}\}$;
- hence $|||e||| \le \|\mathrm{Res}\|_* \le \eta_A + \eta_S$, $|||e|||_\oplus \le \kappa(\eta_A + \eta_S)$, and $E \le \eta_E$.

Proof.

1. For any $\sigma_h \in H(\mathrm{div})$ with $\sigma_h \cdot n = -\Pi_F g_\ell$ on Γ_N: $\ell(v) = (r_\ell - \Pi_1 r_\ell, v) + (\Pi_1 r_\ell - \mathrm{div}\,\sigma_h, v) - (\tau, \nabla v) + (g_\ell - \Pi_F g_\ell, v)_{\Gamma_N}$.
2. By construction $\Sigma_a g\tilde{}_a = \Pi_1 r_\ell - \psi_{\mathrm{def}}$, so $\Pi_1 r_\ell - \mathrm{div}\,\sigma_h = \psi_{\mathrm{def}} - \Sigma_a m^a$.
3. Bound each term:
   - $(\tau, \nabla v)_K$: either Cauchy–Schwarz, or the identity $(\tau, \nabla v)_K = (\tau \cdot n, v - \bar{v}_K)_{\partial K} - (\mathrm{div}\,\tau, v - \bar{v}_K)_K$ together with the trace inequality $\|w\|^2_F \le (|F|/|K|)(\|w\|^2 + (2/d)\, h_K \|w\|\|\nabla w\|)$ and Poincaré;
   - $(\Sigma_a m^a, v)_K \le \sigma^{-1/2}\|\Sigma_a m^a\|_K\, |||v|||_K$;
   - the oscillation term, by Poincaré with $\min\{h/(\pi\varepsilon^{1/2}), \sigma^{-1/2}\}$;
   - (ψ_{def}, v): write $\psi_{\mathrm{def}} = \mathrm{div}\, s + z$, then $(\psi_{\mathrm{def}}, v) = -(s, \nabla v) + (z, v) \le (\varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|z\|^2)^{1/2}\, |||v|||$.
4. Remark 1.2 gives the rest. ∎

Full details are in sv_convection_extension.md §4(i).

Theorem 2 (efficiency). Assume (H1)–(H6) and let $\ell \in \{\mathrm{Res}_A, -S\tilde{}\}$. Define:

- $\omega\tilde{}_K$ = the union of the patches ω_a with $a \in V_K$, and their neighbours;
- $\mathrm{osc}_K(\ell) = \alpha_K \|r_\ell - \Pi_1 r_\ell\|_K + \Sigma_{F \subset \partial K \cap \Gamma_N} \varepsilon^{-1/4} \alpha_F^{1/2} \|g_\ell - \Pi_F g_\ell\|_F$.

Then:

- **local:** $\eta^2_{\ell,K} \le C \Sigma_{K' \subset \omega\tilde{}_K} [\|\ell\|^2_{*,\omega\tilde{}_{K'}} + \mathrm{osc}_{K'}(\ell)^2]$ for every K;
- **global:** $\eta_\ell \le C(\|\ell\|_* + \mathrm{osc}(\ell))$.

The constant C depends only on d and ϑ. It is independent of ε, σ, b, c, h, A and u_h.

Corollary 3 (robust two-sided bound). Under (H1)–(H6):

- $\eta_A + \eta_S \le C(\|\mathrm{Res}_A\|_* + \|S\tilde{}\|_* + \mathrm{osc}) \le C(\max\{1, c_b\}|||e|||_\oplus + 2\|S\tilde{}\|_* + \mathrm{osc})$;
- hence $E \le \eta_E \le C' \kappa \max\{1, c_b\}(E + \mathrm{osc}) \le C''(1 + c_b)^2(E + \mathrm{osc})$.

Proof. Use $\|\mathrm{Res}_A\|_* \le \|\mathrm{Res}\|_* + \|S\tilde{}\|_*$, Remark 1.2, and the fact that η_S is at most the equilibrated bound $\eta_{-S\tilde{}}$. The factor $\kappa = 2 + \max\{1, c_b\}$ from η_E multiplies the $\max\{1, c_b\}$ from the efficiency side. ∎

Corollary 3' (weighted measure, linear in c_b; audit G2). Let $M = \max\{1, c_b\}$, and define

- $|||e|||_{\oplus,c} = M|||e||| + \|B_A'(e, \cdot)\|_*$, and $E_c = |||e|||_{\oplus,c} + \|S\tilde{}\|_* = E + (M - 1)|||e|||$;
- $\eta_{E,c} = (M - 1)\eta + \eta_E$, with $\eta \ge |||e|||$ the energy bound (`eta_E_c`).

Then $E_c \le \eta_{E,c} \le C(1 + c_b)(E_c + \mathrm{osc})$.

Proof.

- **Upper bound.** $E_c = (M - 1)|||e||| + E \le (M - 1)\eta + \eta_E$ (Theorem 1).
- **Lower bound.** $\|\mathrm{Res}\|_* \le M|||e||| + \|B_A'(e, \cdot)\|_* = |||e|||_{\oplus,c}$. With Theorem 2, $\eta \le \eta_A + \eta_S + \mathrm{bd} \le C(|||e|||_{\oplus,c} + 2\|S\tilde{}\|_* + \mathrm{osc})$, and $\eta_E \le C\kappa(|||e|||_{\oplus,c} + 2\|S\tilde{}\|_* + \mathrm{osc})$. Hence $\eta_{E,c} \le C(M - 1 + \kappa)(E_c + \mathrm{osc})$, and $M - 1 + \kappa = 2M + 1 \le 3(1 + c_b)$. ∎
- **Remark.** E_c is equivalent to $\|\mathrm{Res}\|_* + \|S\tilde{}\|_*$ with constants 1 and $2M + 1$. This is the natural measure when $\sigma \ll \|c - \tfrac{1}{2}\,\mathrm{div}\, b\|_\infty$. The c_b-free alternative is the energy norm with the full reaction weight (§7).
- **Smoke check** ($c = 1 + (c_{\max} - 1)xy$, $\sigma = 1$, $\varepsilon = 10^{-3}$ and 10^{-6}, $n = 8$ and 16):

c_b	η_E/E	η_{E,c}/E_c
1	3.1–4.3	equal
10	9.7–12.7	13.4–14.7
100	124–222	82–119
1000	1870–3370	717–1202

Both bounds are guaranteed in every run. The effectivity of η_{E,c} is about 1.2c_b; that of η_E grows faster, though well below the worst case c_b². Test: `test_weighted_augmented`.

3. Tools

A1 (inverse inequalities) [SV20, Lemma 2.2]. For τ ∈ RTN₁(K):

- h_K^{1/2}‖τ·n‖_{∂K} ≤ C_∂‖τ‖_K;
- h_K‖div τ‖_K ≤ C_div‖τ‖_K.

Both constants are explicit in d and ϑ.

A2 (patch stability) [SV20, Lemma 4.1]. Let g ∈ P₁(T_a) and let τ be a broken RTN₁ field on T_a, with the compatibility (g, 1)_{ω_a} = ⟨prescribed normal flux⟩ on compatible patches. Then

min{‖τ + v‖_{ω_a} : v ∈ V^a, div v = g} ≤ C sup{(g, w)_{ω_a} − (τ, ∇w)_{ω_a} : w ∈ H¹_†(ω_a), ‖∇w‖ = 1}.

Here H¹_†(ω_a) consists of the functions with mean zero on ω_a (compatible patches), or vanishing on ∂ω_a ∩ Γ_D. (Renamed, audit T1: H¹_ is the local test space of §1.)* With Neumann faces, the version used is the standard affine shift of the prescribed normal flux [EV15]; not re-proved.

A3 (bubbles) [V05, Lemma 3.6]. With b_K the element bubble and b_F^ϑ the face bubble squeezed by ϑ_F = min{ε^{1/2}α_F/h_F, 1}:

- (w, b_K w)_K ≥ c‖w‖²_K and |||b_K w|||_K ≤ C α_K^{-1}‖w‖_K, for w ∈ P₁(K);
- (μ, b_F^ϑ μ)_F ≥ c‖μ‖²_F, |||b_F^ϑ μ|||_{ω_F} ≤ C ε^{1/4}α_F^{-1/2}‖μ‖_F, and ‖b_F^ϑ μ‖_{ω_F} ≤ C ε^{1/4}α_F^{1/2}‖μ‖_F, for μ ∈ P₁(F).

A4 (quasi-interpolant) [variant of V05, Lemma 3.3]. Let I_c v = Σ_{a compatible} v̄_{ω_a} ψ_a, where v̄_ω is the mean over ω. Then |||I_c v||| ≤ C_I |||v||| for v ∈ H¹_D.

- **The σ-part** follows from |v̄_a| ≤ |ω_a|^{-1/2}‖v‖_{ω_a}.
- **The ε-part.** On K, ∇I_c v = Σ_{a∈V_K} (v̄_a − v̄_K)∇ψ_a − Σ_{b free} v̄_b∇ψ_b.
  - The first sum is bounded by Poincaré on the vertex patches of K.
  - The second sum runs over the excluded vertices b. Each such patch has a face on Γ_D, where v = 0. Friedrichs gives |v̄_b| ≤ C h_b^{1-d/2}‖∇v‖_{ω_b}.
- **Comparison with [V05].** Verfürth's set excludes the vertices on Γ_D. Here, in addition, vertices whose patch has an opposite face on Γ_D are excluded; the same argument applies to them.

A5 (commuting projection) [Christiansen–Winther 2008; Licht 2019, mixed boundary conditions — abstract-level, see §0]. There is Π_h : H(div) ∩ {s·n = 0 on Γ_N} → RT₀ with the same boundary condition, such that:

- ‖Π_h s‖ ≤ C_Π ‖s‖;
- div Π_h s = Π₀ div s.

A6 (finite overlap). If ω_1, …, ω_k ⊂ Ω cover each point at most N times, then Σ_j ‖ℓ‖²_{,ω_j} ≤ N ‖ℓ‖²_.

- Proof: pick v_j ∈ H¹_*(ω_j) (extended by zero, so v_j ∈ H¹_D(Ω)) with |||v_j|||_{ω_j} = 1 and ℓ(v_j) = ‖ℓ‖_{*,ω_j} (up to a factor 1 − δ), and set t_j = ℓ(v_j) and v = Σ_j t_j v_j.
- Then ℓ(v) = Σ_j t_j², while |||v|||² ≤ N Σ_j t_j² |||v_j|||² = N Σ_j t_j².
- So ‖ℓ‖_* ≥ (Σ_j t_j²)^{1/2} / √N. ∎

4. Lemmas

Lemma A (◆ the evaluated flux term carries the SV weight). For τ ∈ RTN₁(K) and ω_K = min{1, ϑ_K^{1/2}}:

min{ε^{-1/2}‖τ‖_K, Σ_F c_F‖τ·n‖_F + m_K‖div τ‖_K} ≤ C_A ε^{-1/2} ω_K ‖τ‖_K.

Proof.

1. If ϑ_K ≥ 1, the first entry of the min gives the bound with C_A = 1.

2. If θ_K < 1, then m_K ≤ σ^{-1/2} = (h_K/ε^{1/2}) θ_K (with equality for θ_K ≤ 1/π) (audit T8).
3. Also 1/σ + h_K/(d(σε)^{1/2}) = (h_K²/ε)(θ_K² + θ_K/d) ≤ 2(h_K²/ε)θ_K, and |F|/|K| ≤ C_ϑ h_K^{-1}. So c_F ≤ C h_K^{1/2} ε^{-1/2} θ_K^{1/2}.
4. With A1: Σ_F c_F‖τ·n‖_F + m_K‖div τ‖_K ≤ C ε^{-1/2}(θ_K^{1/2} C_∂ + θ_K C_div)‖τ‖_K ≤ C_A ε^{-1/2} θ_K^{1/2}‖τ‖_K. ∎

Remark A.1 (consistency with the necessity of weights in [SV20]; audit S2). Smears and Vohralík show that weights in the flux construction are necessary for the robustness of any flux-equilibration estimate that does not employ submeshes or a combination of estimators [SV20]. The construction here has no weights in the patch problems, but it is not covered by that statement: the evaluation is a combination of two estimators, min{ε^{-1/2}‖τ‖_K, Σ_F c_F‖τ·n‖_F + m_K‖div τ‖_K}.

- **The second entry carries the weight.** Lemma A shows it is ≤ C ε^{-1/2} θ_K^{1/2}‖τ‖_K, i.e. the SV-weighted flux norm; the first entry covers the diffusion regime.
- **What the weight-free minimiser needs.** The weight is not needed in the construction, because Lemma G case 1 (the best-approximation trick) shows that the unweighted minimiser is already small in the weighted norm ε^{-1}θ_a‖τ^a‖².
- **What removing the min would do.** The robustness would be lost. That is the situation of SV20's necessity statement, and of the non-robust ESV cut-off that gap 6 corrected.
- **Status.** I read SV20's statement in its arXiv abstract and in a search summary; the precise location in the paper is to be checked.

Lemma B (conforming reconstruction). For each vertex a there is v° ∈ V^a with

‖ψ_a Dv_h + v°‖_{ω_a} ≤ C_B h_a^{1/2} J_a,

where J_a² = Σ_{F ∈ F_a} ‖j_F‖²_F, the sum runs over the faces F containing a, and

- j_F = [[Dv_h·n]] on interior faces;
- j_F = Dv_h·n − Π_F g_ℓ on Neumann faces.

Proof.

1. Define v° by its RTN₁ degrees of freedom:
   - interior moments: those of −ψ_a Dv_h on each K;
   - normal moments on interior faces of T_a: the average of the two traces of −ψ_a Dv_h·n;
   - normal moments on ∂ω_a: 0 (there ψ_a = 0), or the prescribed −Π_F(ψ_a g_ℓ) on Neumann faces, or the one-sided trace on Dirichlet faces.
2. Then w = ψ_a Dv_h + v° is a broken RTN₁ field.
   - Its interior degrees of freedom vanish.
   - Its face moments are ±½ψ_a j_F on interior faces and ψ_a j_F on Neumann faces, and 0 otherwise. These are exact, because ψ_a Dv_h·n ∈ P₁(F).
3. Scaling to the reference element gives ‖w‖_K ≤ C h_K^{1/2} Σ_{F⊂∂K} ‖ψ_a j_F‖_F. Use ‖ψ_a‖_∞ = 1. ∎

Lemma G (◆ patch lemma, weight-free construction). Let RES_a² = Σ_{K∈T_a} α_K²‖r_ℓ‖²_K + Σ_{F∈F_a} ε^{-1/2} α_F ‖j_F‖²_F, and let θ_a = ε^{1/2}σ^{-1/2}/h_a. Then

ε^{-1}ω_a²‖τ^a‖²_{ω_a} + σ^{-1}‖m^a‖²_{ω_a} ≤ C(RES_a² + [θ_a < 1] σ^{-1} c_a² |ω_a|).

For σ = 0 the second term on the left and the bracket are absent.

Proof, case 2 (θ_a ≥ 1, or σ = 0: diffusion regime). Here ω_a = 1.

1. By A2 there is v* ∈ V^a with div v* = g̃_a. The shift makes the data compatible, and the constant c_a drops out against mean-zero test functions.
2. The flux satisfies ‖ψ_a Dv_h + v*‖ ≤ C sup over w ∈ H¹_†(ω_a) with ‖∇w‖ = 1 of [(Π₁(ψ_a r_ℓ), w) + Σ_F (ψ_a j_F, w)_F].
3. This follows from elementwise integration by parts, using div(ψ_a Dv_h)|_K = ∇ψ_a·Dv_h (Dv_h is constant on K).
4. Poincaré on ω_a (Friedrichs on Dirichlet patches) and the trace inequality give ‖ψ_a Dv_h + v*‖ ≤ C(h_a‖r_ℓ‖_{ω_a} + h_a^{1/2}J_a).

5. Hence J_a(σ^a) ≤ J_a(v*) = ε^{-1}‖ψ_a Dᵛu_h + v*‖² ≤ C(ε^{-1}h_a²‖r_ℓ‖² + ε^{-1}h_a J_a²).
6. Since θ ≥ 1, α_K ≃ h_a ε^{-1/2} and α_F ≃ h_a ε^{-1/2}. So the right-hand side is ≤ C RES_a², and the left-hand side of the lemma is ≤ J_a(σ^a).
7. For σ = 0 the same competitor is admissible in the constrained problem. ∎

Proof, case 1 (θ_a < 1: reaction regime; ◆ the best-approximation trick).

1. **Why the SV competitor does not suffice.** Its flux part costs ε^{-1}h_aJ_a², which is θ_a^{-1} times larger than the efficient size ε^{-1/2}σ^{-1/2}J_a². SV20 compensate with the weight w_a² ≃ θ_a in the construction. Here the weight is only in the evaluation.
2. **The comparison flux.** Let v_b = argmin over v ∈ V^a of ‖ψ_a Dᵛu_h + v‖, with no divergence condition, and let δ = ‖ψ_a Dᵛu_h + v_b‖ (this δ is an approximation error, not the defect δ_a). Since σ^a ∈ V^a, ‖τ^a‖ ≥ δ. Optimality J_a(σ^a) ≤ J_a(v_b) gives ε^{-1}‖τ^a‖² + σ^{-1}‖m^a‖² ≤ ε^{-1}δ² + σ^{-1}‖m_b‖², with m_b = div v_b − g˜_a.
3. **Equilibration term:** σ^{-1}‖m^a‖² ≤ σ^{-1}‖m_b‖². This uses δ ≤ ‖τ^a‖.
4. **Bound on m_b.**
   - Write m_b = div_h(ψ_a Dᵛu_h + v_b) − Π₁(ψ_a r_ℓ) + c_a.
   - A1 gives ‖div_h(…)‖ ≤ C h_a^{-1}δ, and Lemma B gives δ ≤ ‖ψ_a Dᵛu_h + v°‖ ≤ C h_a^{1/2}J_a.
   - So σ^{-1}‖m_b‖² ≤ C(σ^{-1}h_a^{-1}J_a² + σ^{-1}‖r_ℓ‖²_{ω_a} + σ^{-1}c_a²|ω_a|).
   - Since θ_a < 1, α_F ≃ σ^{-1/2} and α_K ≃ σ^{-1/2}. So σ^{-1}h_a^{-1}J_a² = θ_a ε^{-1/2}σ^{-1/2}J_a² ≤ C ε^{-1/2}α_FJ_a², and σ^{-1}‖r_ℓ‖² ≤ C Σ α_K²‖r_ℓ‖².
5. **Flux term.**
   - From step 2, ‖τ^a‖² ≤ δ² + (ε/σ)‖m_b‖².
   - Multiply by ε^{-1}ω_a² = ε^{-1}θ_a: ε^{-1}θ_a‖τ^a‖² ≤ θ_a ε^{-1}δ² + θ_a σ^{-1}‖m_b‖².
   - The first term is ≤ C θ_a ε^{-1}h_aJ_a² = C ε^{-1/2}σ^{-1/2}J_a². The second is θ_a times the bound of step 4.
6. Both terms are therefore ≤ C(RES_a² + σ^{-1}c_a²|ω_a|). ∎

Lemma H (defect terms).

(i) **Local defect term.** For θ_a < 1: σ^{-1}c_a²|ω_a| ≤ C‖ℓ‖²_{*,ω_a}.

- Proof: ψ_a ∈ H¹_*(ω_a) for compatible a (it vanishes on ∂ω_a \ Γ_N), so c_a|ω_a| = ℓ(ψ_a) ≤ ‖ℓ‖_{*,ω_a}|||ψ_a|||_{ω_a}, and |||ψ_a|||²_{ω_a} ≤ C(εh_a^{-2} + σ)|ω_a| = Cσ(θ_a² + 1)|ω_a| ≤ 2Cσ|ω_a|. ∎

(ii) **Global defect term** (◆). The functional ℓ_def(v) = (ψ_def, v) satisfies ‖ℓ_def‖_* ≤ C_I‖ℓ‖_*. If A5 holds, then also η_{ℓ,def} ≤ C_Π C_I ‖ℓ‖_*.

- Proof.
  1. ℓ_def(v) = Σ_{a compatible} c_a(χ_{ω_a}, v) = Σ_a ℓ(ψ_a) v̄_{ω_a} = ℓ(I_c v). A4 gives the first claim.
  2. For the second, let (s, z) attain the continuous dual norm: ψ_def = div s + z, with ‖ℓ_def‖²_* = ε^{-1}‖s‖² + σ^{-1}‖z‖². This is the duality for the symmetric operator −εΔ + σ, and z = 0 if σ = 0.
  3. Then s_h = Π_h s is admissible for the discrete minimum, because div s_h = Π₀ div s = ψ_def − Π₀z (ψ_def ∈ P₀).
  4. Hence η²_{ℓ,def} ≤ ε^{-1}C_Π²‖s‖² + σ^{-1}‖z‖² ≤ C_Π²‖ℓ_def‖²_*. ∎

(iii) **For ℓ = Res_A,** ℓ(ψ_a) = δ_a^alg + δ_a^quad (§1), i.e. minus the algebraic residual at a up to the quadrature defect. So at the discrete solution both defect terms reduce to the quadrature part: they are bounded by C osc_q, and vanish when the data are integrated exactly (corrected, audit R1). At an iterate they are bounded by ‖Res_A‖_*. This makes Theorem 2 hold for inexact iterates.

Lemma D (residual estimator ≤ local dual norm) [V05 proof of Thm. 4.1, with A3]. For every K and every face F:

- α_K‖Π₁r_ℓ‖_K ≤ C(‖ℓ‖_{*,K} + α_K‖r_ℓ − Π₁r_ℓ‖_K);
- ε^{-1/4}α_F^{1/2}‖j_F‖_F ≤ C(‖ℓ‖_{*,ω_F} + α_F‖r_ℓ‖_{ω_F} + Neumann data oscillation).

Proof.

1. **Element term.** $\ell(b_K w) = (r_\ell, b_K w)_K$ for $w \in P_1(K)$. The diffusion term vanishes, because $b_K = 0$ on ∂K and $\operatorname{div}(D\nabla u_h) = 0$ in K. Take $w = \Pi_1 r_\ell$ and apply A3.
2. **Face term.** $\ell(b_F^\theta \mu) = (r_\ell, b_F^\theta \mu)_{\omega_F} + (j_F, b_F^\theta \mu)_F$. Take $\mu = j_F$, apply A3, and use the element bound. On a Neumann face, $b_F^\theta \mu \in H^1_*(\omega_F)$ does not vanish on F, and the Neumann data oscillation enters.
3. Only the form of ℓ is used; the bilinear form B and u_h's equation are not. ∎

5. Proof of Theorem 2

1. **Lemma A on each cell.** On K, $\tau = \sum_{a\in V_K} \tau^a$. With $\omega_K \simeq \omega_a$ for $a \in V_K$ (shape regularity): $\eta_{\ell,K} \le C \sum_{a\in V_K} [\varepsilon^{-1/2}\omega_a\|\tau^a\|_K + \sigma^{-1/2}\|m^a\|_K] + m_K\|r_\ell - \Pi_1 r_\ell\|_K$. The last term uses $m_K \le \alpha_K$.
2. **Lemma G for each a,** then Lemma H(i) for the local defect: $\eta^2_{\ell,K} \le C \sum_{a\in V_K} (\mathrm{RES}_a^2 + \|\ell\|^2_{*,\omega_a}) + \mathrm{osc}_K^2$.
3. **Lemma D,** with $\|r_\ell\| \le \|\Pi_1 r_\ell\| + \|r_\ell - \Pi_1 r_\ell\|$: $\mathrm{RES}_a^2 \le C \sum_{K'\subset\omega_a, F\in F_a} (\|\ell\|^2_{\omega_F} + \|\ell\|^2_{K'} + \mathrm{osc}^2_{K'})$. This is the local bound.
4. **Summing over K.** The sets K, ω_F and ω_a overlap at most $N(d, \vartheta)$ times. A6 gives $\sum_K \eta^2_{\ell,K} \le C(\|\ell\|^2_* + \mathrm{osc}^2)$. Adding Lemma H(ii) for $\eta_{\ell,\mathrm{def}}$ gives the global bound. ∎

Remark 5.1 (where robustness comes from). Every constant above depends only on d and ϑ. The parameters ε, σ, b and c enter only through:

- the weights α and θ, which cancel exactly in Lemmas A, G and H(i);
- the functional ℓ itself (through D, r_ℓ and g_ℓ), which is never bounded by the coefficients.

The stabilisation tensor A appears only inside ℓ (as D). It therefore does not affect the constants, whatever the limiter.

Remark 5.2 (local versus global). Theorem 2 is local in the dual norms of ℓ. The conversion to the error is local only for $|||e|||_{\omega^\sim_K}$. The convective part $\|B_{A'}(e, \cdot)\|_{*,\omega^\sim_K}$ is a localised dual norm, as in [V05]. Global efficiency in E is Corollary 3.

Remark 5.3 (the direct version). The `direct` variant ($D = \varepsilon I$, $\sigma > 0$) is not compatible in general: $\ell(\psi_a) = \mathrm{Res}(\psi_a) = S(u_h; u_h, \psi_a)$ at the discrete solution, up to the quadrature defect of §1 (audit T2). Its efficiency would carry the consistency term $\sum_a \sigma^{-1} S(u_h; u_h, \psi_a)^2/|\omega_a|$. Corollary 3 does not need it, because galerkit uses $\min\{\eta_{\mathrm{dir}}, \eta_A + \eta_S\} \le \eta_A + \eta_S$.

Remark 5.4 (where A5 is needed). A5 is used in Lemma H(ii), the quasi-optimality of the global RT_0 defect bound, and through it in the bound on η_{alg} in Proposition 5.2 (audit T6). That covers both $\sigma > 0$ and $\sigma = 0$, where $z = 0$. The discrete inf-sup condition alone gives $\|s_h\|_{H(\mathrm{div})} \lesssim \|\psi_{\mathrm{def}}\|$, which is not $\|s_h\| \lesssim \|\psi_{\mathrm{def}}\|_{H^{-1}}$.

- **$\Gamma_N = \emptyset$:** the space has no boundary condition, and the smoothed projection of [CW08] suffices.
- **Mixed boundary conditions:** these need the version of Licht (2019) (abstract-level, see §0; audit T3).
- **What it affects:** only the defect terms. For $\ell = \mathrm{Res}_A$ at the discrete solution they reduce to the quadrature part of §1 ($\|\ell_{\mathrm{def}}\|_* = \mathrm{osc}_q$; zero for exactly integrated data) (audit T2). Without A5, Theorem 2 still holds for $\ell = \mathrm{Res}_A$ at the discrete solution, with $C_\Pi \mathrm{osc}_q$ replaced by the computed $\eta_{\mathrm{def},A}$. For $\ell = -S^\sim$, the defect term then has only the upper bound $\|\ell_{\mathrm{def}}\|_* \le C_I\|S^\sim\|_*$ for the true dual norm, not for its RT_0 computation.

6. Numerical verification (smoke runs)

Lemmas A and G with Prop. 2 ($\eta \lesssim$ local residual estimator). For the outflow-layers problem, both functionals, BJK and SMUAS, on uniform meshes:

	quantity	max over $\sigma \in \{0, 10^{-2}, 1\}$, $\varepsilon = 10^{-2} \dots 10^{-8}$, $n \in \{8, 32\}$
$\ell = \mathrm{Res}_A$	$\max_K \eta^2_{\ell,K} / \sum_{K'\sim K} (\mathrm{RES}^2_{K'} + \mathrm{osc}^2_{K'})$	0.63
$\ell = \mathrm{Res}_A$	$\sum_K \eta^2_{\ell,K} / \sum_K (\mathrm{RES}^2_K + \mathrm{osc}^2_K)$	2.39
$\ell = -S^\sim$	$\max_K$	0.56

	quantity	max over σ ∈ {0, 10^{-2}, 1}, ε = 10^{-2} … 10^{-8}, n ∈ {8, 32}
$\ell = -S^{\sim}$	global	2.36

- **No growth in ε.** The ratios rise from about 0.03 (diffusion regime) to about 2.4 as the reaction regime sets in, and then stay bounded.
- **Lemma D for $\ell = -S^{\sim}$.** Here a reference is available: the P2 Riesz representative, a lower bound of $\|S^{\sim}\|_*$. With it, RES/$\|S^{\sim}\|_*$ = 2.4–2.5 for σ = 0 and ε = 10^{-2} … 10^{-8}.
  - For σ = 1 the ratio grows (3.9 at ε = 10^{-6}, 12 at 10^{-8}). This is the known reference artifact: $\|S^{\sim}\|_*$ concentrates in layers of width √(ε/σ) that the same-mesh P2 space cannot resolve, as the nested-mesh check of gap 6 showed.
- **Corollary 3.** The effectivities of `sv` in E are 2.9–3.8 for σ = 0 and 3.2–10 for σ = 1, against same-mesh references (§9 of the summary). The upper end for σ = 1 is the same artifact.

Scripts. `scratchpad/g9.py` (ratios) and `tests/test_cdr_sv_estimator.py` (bounds).

7. What is not covered

- **Variable ε.** Reliability and local efficiency are now in §11. Robustness in coefficient jumps is shown only numerically, and only for quasi-monotone ε.
- **Non-P1 Dirichlet data.** The efficiency of the gap 4 lifting terms is not treated.
- **Neumann faces in A2.** The affine-shift version of SV20 Lemma 4.1 is cited, not re-proved.
- **A5 with mixed boundary conditions.** Confirmed at abstract level only (Licht 2019); needed for σ > 0 and σ = 0 alike (audit T3, T6).
- **The dependence on c_b in Corollary 3.** It is $(1 + c_b)^2$ in E and $(1 + c_b)$ in the weighted measure E_c (Corollary 3'). Removing it altogether needs the energy norm with the full reaction $c - \frac{1}{2}\,\mathrm{div}\, b$, which galerkit's σ only bounds from below.
- **The direct variant (Remark 5.3)** and the gap 6 hybrid have no efficiency claim.

8. References

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- **[AV14]** M. Ainsworth, T. Vejchodský, Robust error bounds for finite element approximation of reaction–diffusion problems with non-constant reaction coefficient in arbitrary space dimension, CMAME 281 (2014) 184–199, doi 10.1016/j.cma.2014.08.005, arXiv 1401.2394: extends [AV11] to arbitrary dimension, mixed boundary conditions and a **piecewise-constant** reaction coefficient. Fully variable reaction was still listed as open by Vejchodský in 2018. [abstract]

9. Addendum: local efficiency in a localisable error measure (gap 4 of the scan)

The obstacle and the answer

The obstacle. Theorem 2 is local in the dual norms of the residual functionals. Converting them to the error brings in ‖B_A'(e, ·)‖_*, and this global dual norm is not a sum of local contributions.

The answer. Everything in E localises except one term. That term depends only on u_h, and the estimator already computes it. The tool is the localisation of dual norms of functionals that vanish on a partition of unity [BMV].

Definitions

Local dual norm. For each vertex a, $H^1_*(\omega_a) = \{v \in H^1(\omega_a) : v = 0 \text{ on } \partial\omega_a \setminus \Gamma_N\}$, and $\|\ell\|_{,a} = \sup\{\ell(v)/|||v|||_{\omega_a} : v \in H^1_(\omega_a)\}$.

Localised error measure.

$E_{loc}^2 = |||e|||^2 + \Sigma_a (\|B_A'(e, \cdot)\|^2_{,a} + \|S^\sim\|^2_{,a})$.

The local form is $E_{loc}(\omega)^2$, with the sums restricted to the patches in ω. E_loc is a sum of local contributions by construction. Its first two parts form the localised augmented norm $|||e|||^2_{\oplus,loc} = |||e|||^2 + \Sigma_a \|B_A'(e, \cdot)\|^2_{*,a}$.

Nodal moment functionals. For $\ell \in \{\mathrm{Res}_A, S^\sim\}$, $\ell_{def}(v) = \Sigma_{a \text{ compatible}} \ell(\psi_a)\, \bar{v}_{\omega_a}$.

- For $S^\sim$, $\ell(\psi_a) = S(u_h; u_h, \psi_a)$: the nodal moments of the stabilisation.
- For Res_A, $\ell(\psi_a)$ is minus the algebraic residual up to the quadrature defect of §1, so it reduces to δ^{quad} at the discrete solution.
- Their dual norms are bounded above by the computed defect terms $\eta_{def,S}$ and $\eta_{def,A}$ (Theorem 1, step 3). With A5 they are also bounded below by them (Lemma H(ii)).

Results

Lemma L (robust localisation). For every $\ell \in (H^1_D)'$, $\Sigma_a \|\ell\|^2_{,a} \le (d+1)\|\ell\|^2_$. For $\ell \in \{\mathrm{Res}_A, S^\sim\}$ (or any ℓ with the nodal moments $\ell(\psi_a)$ of §1), also

$\|\ell\|_* \le C_L (\Sigma_a \|\ell\|^2_{,a})^{1/2} + \|\ell_{def}\|_$.

(Scope stated, audit T4: Theorem 4.1 applies the first inequality to B_A'(e, ·).)

C_L depends only on d and ϑ. It does not depend on ε or σ.

Proof.

1. Let $v \in H^1_D$. Set $\bar{v}_a$ = mean over ω_a of v for compatible a, and $\bar{v}_a = 0$ for patches with a free (Dirichlet) face. Since $\Sigma_a \psi_a = 1$, $\ell(v) = \Sigma_a \ell(\psi_a(v - \bar{v}_a)) + \Sigma_{a \text{ compatible}} \bar{v}_a \ell(\psi_a) = \Sigma_a \ell(\psi_a(v - \bar{v}_a)) + \ell_{def}(v)$, and $\psi_a(v - \bar{v}_a) \in H^1_*(\omega_a)$.
2. **The robust bound $|||\psi_a(v - \bar{v}_a)|||_{\omega_a} \le C|||v|||_{\omega_a}$.**
   - Diffusion part: $\varepsilon\|\nabla(\psi_a w)\|^2 \le 2\varepsilon(\|\nabla w\|^2 + \|\nabla\psi_a\|^2_\infty \|w\|^2)$, with $w = v - \bar{v}_a$. Poincaré (compatible patches) or Friedrichs (patches with a Dirichlet face) gives $\|w\| \le C\, h_a \|\nabla v\|_{\omega_a}$, and $\|\nabla\psi_a\|_\infty \le C\, h_a^{-1}$.
   - Reaction part: $\sigma\|\psi_a w\|^2 \le 2\sigma(\|v\|^2 + |\omega_a|\, \bar{v}_a^2) \le 4\sigma\|v\|^2_{\omega_a}$.
3. Hence $|\Sigma_a \ell(\psi_a(v - \bar{v}_a))| \le C(\Sigma_a \|\ell\|^2_{,a})^{1/2}(\Sigma_a |||v|||^2_{\omega_a})^{1/2} \le C\sqrt{(d+1)}(\Sigma_a \|\ell\|^2_{,a})^{1/2}|||v|||$.
4. The second inequality is A6 with N = d+1, the maximal number of vertex patches covering a point. ∎

Theorem 4 (localisable two-sided bounds). Under (H1)–(H6). In parts 2 and 3, c_b denotes the σ-based constant $c_{b,\sigma} := \|c - ½ \operatorname{div} b\|_\infty / \sigma$ ($\sigma > 0$), or $c_{b,\sigma} := 0$ if $c = ½ \operatorname{div} b$; the Friedrichs-based entry of (H3) is not admissible there, because it is a global constant (corrected, audit T5). For $\sigma = 0$ with $c \ne ½ \operatorname{div} b$, see Remark 9.3.

1. **Upper bound, guaranteed:** $E_{loc} \le \sqrt{(d+1)}\, E \le \sqrt{(d+1)}\, \eta_E$ (`eta_Eloc`).
2. **Local efficiency, robust:** $\eta_{A,K} + \eta^{patch}_{S,K} \le C(1 + c_b)(E_{loc}(\omega^\sim_K) + \mathrm{osc}(\omega^\sim_K))$. Here $\eta^{patch}_{S,K}$ is the cellwise part, without the global defect term.
3. **Equivalence modulo computable terms:** $E \le C(1 + c_b)^2(E_{loc} + \eta_{def,S} + \eta_{def,A})$. At the discrete solution, $\eta_{def,A}$ reduces to the quadrature part: $\|\ell_{def}\|_* = \mathrm{osc}_q$, and with A5 also $\eta_{def,A} \le C_\Pi\, \mathrm{osc}_q$; both are zero for exactly integrated data (audit R1, S1). The bound itself needs no A5, since it uses only $\|\ell_{def}\|_* \le \eta_{def}$. (Corrected, audit G2: the constant is $\kappa \max\{1, c_b\}$.)

Together: E_loc ≲ E ≲ E_loc + η_{def,S} + η_{def,A}, where η_{def,A} reduces to the quadrature part at the discrete solution (audit T2). In the global error measure, the only non-localisable part is the dual norm of the stabilisation's nodal moments. It is a function of u_h alone, and the estimator computes it.

Proof.

1. **Part 1** is Lemma L (second inequality) applied to $\ell = B_A'(e, \cdot)$ and $\ell = S^\sim$, plus Theorem 1. The functional B_A'(e, ·) has no partition-of-unity property, so only this direction holds for it.
2. **Part 2.** The local bounds of Theorem 2 use $\|\ell\|_{*,K}$ and $\|\ell\|_{*,\omega_F}$, both of which are ≤ $\|\ell\|_{*,a}$ for a ∈ K or a ∈ F. Then:
   - for ℓ = Res_A: $\|Res_A\|_{*,a} \le \|Res\|_{*,a} + \|S^\sim\|_{*,a}$;
   - $\|Res\|_{*,a} \le \max\{1, c_{b,\sigma}\}|||e|||_{\omega_a} + \|B_A'(e, \cdot)\|_{*,a}$, because B = B_S' + B_A' and $|B_S'(e, v)| \le \max\{1, c_{b,\sigma}\}|||e|||_{\omega_a}|||v|||_{\omega_a}$ for v ∈ H¹_*(ω_a). This uses $\|e\|_{\omega_a} \le \sigma^{-1/2}|||e|||_{\omega_a}$; a Friedrichs inequality is not available for e on a patch, because e does not vanish on ∂ω_a (audit T5);
   - for $\ell = S^\sim$, the local defect term of Lemma H(i) is ≤ $C\|S^\sim\|_{*,a}$.
3. **Part 3.**
   - First, $E \le \kappa\|Res\|_* + \|S^\sim\|_* \le \kappa\|Res_A\|_* + (\kappa + 1)\|S^\sim\|_*$.
   - Apply the first inequality of Lemma L to Res_A and to S^~, then bound $\|Res_A\|_{*,a}$ as in part 2.
   - Finally use $\|\ell_{def}\|_* \le \eta_{def}$. This direction needs no A5. ∎

Remarks

Remark 9.1 (is E_loc weaker than E?). Yes, by part 1.

- **Example where it is much weaker.** Take B_A'(e, ·) smooth and of global extent, with σ = 0. Then $\|B_A'(e, \cdot)\|_* \sim \varepsilon^{-1/2}\|b\cdot\nabla e\|_{H^{-1}}$, while $\Sigma_a \|\cdot\|^2_{*,a} \sim \varepsilon^{-1}h^2\|b\cdot\nabla e\|^2$.
- **Why E is still controlled.** Part 3 says this can only happen when $\|Res_A\|_*$ and $\|S^\sim\|_*$ are themselves globally large. For Res_A that is excluded by Lemma L. So it can only come from the stabilisation's nodal moments, i.e. η_{def,S}, which is computed.
- **In the smoke runs,** η_{def,S}/E_loc = 0.08–0.54 and E_loc/E = 0.48–1.10.

Remark 9.2 (literature).

- The localisation argument is that of [BMV] (W^{-1,q}, partition of unity), in a robust weighted form.
- Verfürth [V05] and Cai & Cai prove efficiency only globally in the augmented norm. Sangalli (Math. Comp. 2008) uses a different, almost-robust norm (1D).
- I found no result that localises the augmented norm for stabilised schemes, or that isolates the stabilisation's nodal moments as the only global term. This is based on searches of abstracts only.

Remark 9.3 (σ = 0 with c ≠ ½ div b; audit T5). For v ∈ H¹_*(ω_a), v vanishes on ∂ω_a \ Γ_N, which has positive measure, so the local Friedrichs inequality $\|v\|_{\omega_a} \le C_P h_a \|\nabla v\|_{\omega_a}$ holds with C_P = C_P(d, ϑ). Hence

((c − ½ div b) e, v)_{ω_a}	≤ μ_a			v		

and $\|Res\|_{*,a} \le |||e|||_{\omega_a} + \mu_a + \|B_A'(e, \cdot)\|_{*,a}$. Part 2 of Theorem 4 therefore holds for σ = 0 with c_{b,σ} replaced by 0 and E_loc² augmented by $\Sigma_a \mu_a^2$. Part 3 holds with the same augmentation and the constant Cκ, with κ = 2 + max{1, c_b} from (H3); the Friedrichs-based c_b is admissible there because the first step, $E \le \kappa\|Res_A\|_* + (\kappa + 1)\|S^\sim\|_*$, is global. The extra term is a weighted L² error, of higher order where h_a ≪ ε^{1/2}. The guaranteed upper bound of part 1 is not affected.

Numerics (smoke runs; same-mesh patch P2 references; `localized_error`)

	σ = 0, ε = 10⁻² … 10⁻⁸	σ = 1, ε = 10⁻² … 10⁻⁸
E_loc / E	0.48–1.05	0.52–1.10
η_E / E_loc	3.6–5.9	3.1–11
max_K (η²_{A,K} + η²_{S,K}) / E_loc(ω~_K)²	0.09–0.64	0.19–49

- **σ = 0:** local efficiency holds, with ratios ≤ 0.64, flat in ε.

- **σ = 1, small ε:** the large local ratios are the unresolved reference. Recomputing the patch dual norms on k-times refined patches (outflow layers, $\varepsilon = 10^{-6}$, n = 8) lowers the ratio from 4.81 (k = 1) to 3.00, 1.85 and 1.14 (k = 2, 4, 8). For $\varepsilon = 10^{-8}$ it falls from 49 to 9.4 at k = 8, still decreasing; layers of width 10^{-4} are not yet resolved there. For σ = 0 the refined reference converges, at 0.09.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_localized_measure`.

Reference added.

- [BMV] J. Blechta, J. Málek, M. Vohralík, Localization of the W^{-1,q} norm for local a posteriori efficiency, HAL hal-01332481; published in IMA J. Numer. Anal. 40 (2020) [from memory — verify].

10. Addendum: adaptive stopping of the nonlinear iteration (item 6 of the scan)

Setting. u^i is any iterate of any solver for the algebraically stabilised system: fixed point, Newton, or a truncated linear solve. Write E(u^i) for the measure E evaluated at u^i, with S~ the stabilisation functional of the iterate.

The split. At an iterate, Res_A(ψ_a) = −r_a + δ_a^quad, where r is the assembled algebraic residual vector (§1). The A-version moves r into the defect shift, so the A-version bound splits as κη_A + η_S + bd = η_disc + η_alg:

- η_alg := κ · (the RT_0 bound of the spread assembled residual Σ_a δ_a^alg χ_{ω_a}/|ω_a|), computed by `sv_algebraic` from r alone. It equals κη_{def,A}(u^i) when the assembly integrates all data exactly (δ^quad = 0, §1; audit N3, R1);
- η_disc := κη_A + η_S + bd − η_alg.

Proposition 5 (corrected, audit G4: the earlier part 3 assumed $C_2\gamma < 1$, which is not needed).

1. **Reliability at every iterate:** E(u^i) ≤ η_E(u^i) ≤ η_disc + η_alg. No assumption is made on the solver.
2. **Efficiency at every iterate.** Assume A1–A5. Then, for every iterate of every solver and every γ, η_disc + η_alg ≤ C(1 + c_b)²(E(u^i) + osc(u^i)) and η_alg ≤ κ C_Π (C_I ‖Res_A(u^i)‖_* + osc_q), where osc_q is the dual norm of the spread quadrature defect of §1 (zero for exactly integrated data; audit N3, R1). For the weighted measure, η_{E,c}(u^i) ≤ C(1 + c_b)(E_c(u^i) + osc(u^i)). The constants are independent of ε, σ, h and the iteration index.
3. **What the stopping rule adds.** If the iteration stops at η_alg ≤ γ η_disc, then E(u^i) ≤ (1 + γ)η_disc. The guaranteed bound is then carried by η_disc up to the factor 1 + γ: the algebraic part is at most the fraction γ/(1 + γ) of the current bound. (Corrected, audit N1: this says nothing about later iterates, whose η_disc changes as well.)

Proof.

- **Part 1** is §1 (reliability) applied with the shifted data. The shift is admissible on compatible patches because Σ_a δ_a ψ_a reproduces Res_A on P1 test functions.
- **Part 2.** Nothing in Theorem 2 uses the discrete equation; Lemma D tests only the form of ℓ. At an iterate it therefore gives η_A(u^i) ≤ C(‖Res_A(u^i)‖_* + osc), including the defect term:
  - the local shift c_a by Lemma H(i);
  - the global term η_{def,A} ≤ C_Π C_I ‖Res_A(u^i)‖_* by Lemma H(ii) with A5. For η_alg, which spreads δ^alg = δ − δ^quad, the same argument gives ‖ℓ^alg_def‖_* ≤ C_I‖Res_A(u^i)‖_* + osc_q, hence the bound on η_alg in part 2.

Further, Res(u^i)(v) = (f, v) + (g_N, v)_{Γ_N} − B(u^i, v) = B(u − u^i, v) for every v ∈ H¹_D. Hence ‖Res_A(u^i)‖_* ≤ max{1, c_b}|||u − u^i|||_⊕ + ‖S~(u^i)‖_* (Remark 1.2), where S~ and A are those of the iterate. The rest is Corollary 3 (Corollary 3' for E_c) at u^i, using η_disc + η_alg = κη_A + η_S + bd.

- **Part 3** follows from part 1. ∎

Hypotheses and sharpness.

- **No condition on γ.** The earlier proof bounded η_disc by Theorem 2 and then absorbed a separate term C(Σ_a ‖ψ_def‖²_{,ω_a})^{1/2} ≤ C_2 η_alg. That term is already part of Theorem 2 (Lemma H(i) bounds it by ‖Res_A(u^i)‖_{,ω_a}), so it was counted twice, and the smallness condition $C_2\gamma < 1$ is superfluous. In Ern–Vohralík 2013 the smallness condition is used to control the linearisation and algebraic estimators by

the discretisation one in the efficiency proof [from memory — verify]. Here it is not needed, because Theorem 2 never uses the discrete equation: the iterate is a conforming function like any other, and E(u^i) contains its algebraic error.

- **A5 is needed only for η_alg ≲ ‖Res_A(u^i)‖_***, i.e. for the global defect. Without A5, η_disc (its patch part) is still efficient, and η_alg is still a guaranteed upper bound of the dual norm of the spread residual.
- **What the rule does not give.** Part 3 is a statement about the iterate. The Ern–Vohralík-type conclusion E(u^i) ≤ C E(u_h), "the stopped iterate is as good as the discrete solution", needs |||u_h − u^i||| ≲ η_alg. That is a discrete stability (inverse) estimate for the nonlinear AFC operator, which is not available without monotonicity. This remains open.
- **Free patches.** On patches with a free (Dirichlet) face, the algebraic residual of a free vertex is absorbed by the patch flux, not by η_alg. This only happens at free vertices next to a Dirichlet face, such as Neumann vertices next to a Dirichlet face. η_disc then contains a small algebraic part, which is harmless for all three parts.
- **Smoke check of part 2** (`scratchpad/g4.py`):
  - setup: smooth and outflow-layers problems, σ = 0 and 1, BJK and SMUAS, ε = 10^{-2} and 10^{-6}, n = 8, iterates i = 1, 2, 5, 20, 100, 2000;
  - η_E(u^i)/E(u^i) stays in 2.9–6.0 at every iterate, including i = 1, where η_alg/η_disc is as large as 0.65;
  - test: `test_iterate_efficiency`.

Literature and novelty.

- **[EV13]** (SISC 35, 2013) is the framework: stopping when η_alg ≤ γ η_disc. There it is built for nonlinear diffusion, with an algebraic flux from ν extra solver steps.
- **Here** the algebraic flux comes from one global RT_0 solve of the spread residual, with no extra iterations. The criterion is stated in the measure E, which is robust in ε.
- **[HPSV21]** (Numer. Math. 147, 2021) proves quasi-optimal cost, but only for strongly monotone Lipschitz operators. The AFC operator is in general not monotone (BJK16 has existence via Brouwer only), so that theory does not apply. Rate-optimal cost for AFC is open.
- **Cheaper algebraic fluxes** exist: the multilevel construction of Papež, Rüde, Vohralík and Wohlmuth (CMAME 2020) is O(N). This would replace the sparse RT_0 solve if that solve becomes the bottleneck in 3D [from memory — verify].
- **AFC-specific stopping.** In my search I found no a posteriori stopping rule for AFC iterations. Ahmed–Jha (arXiv 2602.14504, 2026) relate smoother stabilisations to cheaper iterations; that is based on its abstract, and its full text still has to be checked for a stopping rule.
- **Verdict:** an application of [EV13]. The new parts are the measure E and the iteration-free η_alg.

Numerics (2D smoke runs; BJK, 4 adaptive loops; `stop` = γ):

problem	ε	γ	iterations per loop	E(u^i) at the last loop	η_E/E	η_alg/η_disc at stop
smooth, σ = 0	10^{-2}	— (tol $10^{-8}\sqrt{N}$)	14–26	$1.431\cdot10^{-2}$	3.18–3.46	—
smooth, σ = 0	10^{-2}	0.1	5	$1.392\cdot10^{-2}$	3.29–3.49	0.02–0.04
smooth, σ = 0	10^{-2}	0.01	10	$1.430\cdot10^{-2}$	3.19–3.46	≤ 0.005
smooth, σ = 0	10^{-6}	—	10 000 (not converged)	2.067	3.76–3.99	—
smooth, σ = 0	10^{-6}	0.1	10–15	1.749	3.98–4.53	0.07–0.09
smooth, σ = 0	10^{-6}	0.01	95–1130	2.601	3.81–4.29	≤ 0.009
smooth, σ = 0	10^{-6}	0.001	10 000 (never stops)	2.067	3.76–3.99	0.001–0.004
outflow layers, σ = 0 and 1, BJK and SMUAS	10^{-4}	0.1	5	unchanged	2.8–3.7	≤ 0.03

- **ε = 10^{-6}, γ = 0.1.** The fixed point for BJK does not converge in 10^4 iterations. The stopped iterate after 10–15 iterations has a smaller E than the 10^4-th iterate, and its bound holds.

- **$\gamma = 10^{-3}$.** η_{alg} stalls at 0.1–0.4 % of η_{disc}, so the rule never fires. This is a property of the fixed point, not of the estimator.
- **Effectivities** move by at most 15 %.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_adaptive_stop`, which checks `sv_algebraic` = $\eta_{def,A}$ to four digits and the stopping rule.

References added.

- [EV13] A. Ern, M. Vohralík, Adaptive inexact Newton methods with a posteriori stopping criteria for nonlinear diffusion PDEs, SIAM J. Sci. Comput. 35 (2013) A1761–A1791. HAL [abstract]
- [HPSV21] A. Haberl, D. Praetorius, S. Schimanko, M. Vohralík, Numer. Math. 147 (2021) 679–725. TU Wien repository [abstract]
- [PRVW20] J. Papež, U. Rüde, M. Vohralík, B. Wohlmuth, Sharp algebraic and total a posteriori error bounds for h and p finite elements via a multilevel approach, CMAME 371 (2020). [from memory — verify]

11. Addendum: variable diffusion $\varepsilon = \varepsilon(x)$ (gap 7 extension)

Setting.

- $\varepsilon \in L^\infty(\Omega)$ with $\varepsilon \geq \varepsilon_0 > 0$, and $|||v|||^2 = \|\varepsilon^{1/2}\nabla v\|^2 + \sigma\|v\|^2$.
- $\Pi_1\varepsilon$ is the cellwise L^2 projection of ε onto P1. Its cell mean is $\bar{\varepsilon}_K$.
- $\varepsilon_{lo,K}$ and $\varepsilon_{hi,K}$ are the minimum and maximum of ε on K. In the code (`eps_cell_bounds`), ε is fitted by a P1 function on each cell from its values at the degree-4 points. If the fit is exact, $\varepsilon_{lo,K}$ and $\varepsilon_{hi,K}$ are the extreme vertex values of the fit, which are the true extrema; this covers ε piecewise constant or piecewise P1, also discontinuous across cells. On those cells no other samples are mixed in, so a vertex value taken from the other side of a jump cannot make the bound pessimistic. Otherwise they are sampled extrema (cell mean, vertices, degree-4 and degree-6 points), and the estimator reports the number of such cells in `info`.

The construction. The fluxes use $D = \Pi_1\varepsilon\, I + A$, a P1 tensor per cell, so $D\nabla u_h \in P_1^d \subset RTN_1$. The remainder is the oscillation $\eta_{\varepsilon,K} = \|\varepsilon^{-1/2}(\varepsilon - \Pi_1\varepsilon)\nabla u_h\|_K$.

Patch data (audit G3). $\psi_a D\nabla u_h$ is quadratic, so it is not in RTN_1. The patch problems therefore use its commuting RTN_1 interpolant $\Pi_{RT}(\psi_a D\nabla u_h)$: normal moments against $P_1(F)$, interior moments against P_0^d. Three properties make this work:

- Π_{RT} reproduces RTN_1. So the data are unchanged when $D\nabla u_h$ is constant per cell (constant or piecewise-constant ε, the S-mode).
- $\Sigma_a \Pi_{RT}(\psi_a D\nabla u_h) = \Pi_{RT}(D\nabla u_h) = D\nabla u_h$, so $\tau = D\nabla u_h + \sigma_h = \Sigma_a \tau^a$ exactly, with $\tau^a = \Pi_{RT}(\psi_a D\nabla u_h) + \sigma^a$.
- $\operatorname{div}\Pi_{RT} = \Pi_1 \operatorname{div}$, and $\Pi_{RT}(\cdot)\cdot n|_F = \Pi_F(\cdot)\cdot n$.

The estimator formula and Theorem 1 (which holds for any σ_h) are unchanged. Implemented as `rt1_data` in `sv_estimator`. Against the earlier data the change is 2.6 % for smooth ε at contrast 10^4, with unchanged effectivities.

Lemma V (what P1 tests see). For P1 test functions v, $((\varepsilon - \Pi_1\varepsilon)\nabla u_h, \nabla v)_K = (\nabla u_h\cdot\nabla v)|_K \int_K (\varepsilon - \Pi_1\varepsilon) = 0$, since ∇u_h and ∇v are constant on K and $\Pi_1\varepsilon$ has the cell mean of ε. Three consequences:

- The compatibility $\mathrm{Res}_A(\psi_a) = 0$ at the discrete solution survives with $\Pi_1\varepsilon$ in D, **up to the quadrature defect δ^{quad} of §1** (audit N3, R1). Its diffusion part is $\Sigma_{K\ni a} (Q_K^{asm}(\varepsilon) - Q_K^{est}(\varepsilon))(\nabla u_h\cdot\nabla\psi_a)|_K$, with Q_K the two rules' values of $\int_K \varepsilon$. It vanishes when the degree-4 assembly rule integrates ε exactly (ε of degree ≤ 4 on each cell); the other data terms contribute as in §1.
  - **Reliability** is unaffected: the full δ_a is shifted and its spread is bounded in $\eta_{def,A}$.
  - **The stopping split (Proposition 5)** uses $\eta_{alg} = \kappa\cdot(RT_0 \text{ bound of } \delta^{alg})$ from `sv_algebraic`, i.e. the assembled residual only. The quadrature part stays in $\eta_{disc} = \kappa\eta_A + \eta_S + bd - \eta_{alg}$, so it is not counted as algebraic. In the smoke test with smooth ε (contrast 10^3), $\eta_{def,A} = 1.3\cdot10^{-6}$ at the converged solution, against $2\cdot10^{-11}$ from δ^{alg} alone.
- For any $v \in H^1_D$, $|\mathrm{Res}(v) - \mathrm{Res}_{\{\Pi_1\varepsilon\}}(v)| \leq \Sigma_K \eta_{\varepsilon,K} |||v|||_K$. Reliability therefore holds with $\eta_{\varepsilon,K}$ added cellwise.

- For smooth ε, η_ε,K ≤ C h_K² |ε|_{W^{2,∞}(K)} ε_lo,K^{-1/2} ‖∇u_h‖_K. This is one order above the error, a true oscillation term. A cell-mean ε̄_K would only give O(h)·‖∇ln ε‖_∞ · |||u_h|||_K, which is of the error's own order.

Theorem 6 (variable ε).

1. **Reliability.** Theorem 1 and the upper bounds of Corollaries 3 and 3' hold verbatim, with three changes (wording corrected, audit N2):
   - η_ε,K is added to the A- and direct-version indicators;
   - in every Poincaré, trace and Friedrichs constant (m_K, c_F, the cut-off, the defect solve), ε is replaced by ε_lo,K;
   - ‖ε^{-1/2}τ‖_K is evaluated with ε pointwise. Nothing beyond ε ∈ L^∞ and ε ≥ ε₀ > 0 is assumed for the bound itself. The computed bound is guaranteed when ε_lo,K and ε_hi,K are the true cell extrema: in the code, when ε is piecewise P1 on the mesh. For other ε they are sampled; the bound then holds up to that sampling, and `info` says so.
2. **Local efficiency.** Theorem 2 holds with constants that also depend on the patch contrast κ_ε(ω̃_K) = max_{ω̃_K} ε / min_{ω̃_K} ε, plus the oscillation η_ε. The element residual becomes r̂_ℓ = r_ℓ + div_h(Dψ_a ∇u_h), with div_h(Dψ_a ∇u_h) = ∇Π₁ε·∇u_h ∈ P₀. For ε piecewise constant on the mesh, the within-cell ratio ε_hi,K/ε_lo,K is 1, and only the contrast between cells of a patch remains.

Proof.

- **Part 1.** Lemma V; each Cauchy–Schwarz step weighted by ε pointwise; the inverse inequalities with ε_lo,K, which only make the constants larger.
- **Part 2** (corrected, audit G3).
  - **Lemma B** holds verbatim with the data Π_RT(ψ_a D∇u_h). The field w = Π_RT(ψ_a D∇u_h) + v° is broken RTN₁, its face moments are ±½Π_F(ψ_a j_F), and its interior moments vanish. Before, the uninterpolated ψ_a D∇u_h left a first-order remainder ‖(I − Π_RT)(ψ_a D∇u_h)‖_K ≲ h_K‖∇Π₁ε‖_{∞,K}‖∇u_h‖_K.
  - **A2 / Lemma G** need broken RTN₁ data, which this now is. The integration by parts in case 2, step 3 reads (g̃_a, w) − (Π_RT(ψ_a D∇u_h), ∇w) = (Π₁(ψ_a r̂_ℓ), w) + Σ_F (Π_F(ψ_a j_F), w)_F for mean-zero w, because div_h Π_RT(ψ_a D∇u_h) = ∇ψ_a·D∇u_h + Π₁(ψ_a div_h(D∇u_h)). Case 1 is unchanged.
  - **Lemma D**: ℓ(b_K w) = (r̂_ℓ, b_K w)_K, since div(D∇u_h) ≠ 0 now, and the bubble argument is unchanged. Since div_h(Dψ_a ∇u_h) ∈ P₀, r̂_ℓ − Π₁r̂_ℓ = r_ℓ − Π₁r_ℓ, so the oscillation is unchanged.
  - **Weights.** In Lemmas A and G, the patch weight ε_a, the cell weights ε_K and the norms of the competitor are compared using max/min of ε on ω̃_K. In the bubble estimates, ε appears only through the local energy norms, at the price of the same ratio. ∎

Minimality and sharpness.

- **Reliability is minimal:** ε ∈ L^∞ with ε ≥ ε₀ is what the norm needs anyway.
- **Efficiency is not robust in jumps as proved.** Removing the κ_ε dependence needs quasi-monotonicity of ε, as in Bernardi–Verfürth 2000 and Petzoldt 2002 for residual estimators and in the Cai et al. equilibrated estimators. It also needs ε-harmonic patch weights in the minimisation.
- **Without quasi-monotonicity it is impossible.** Li (arXiv 2609.27817, 2026) shows that no locally computable estimator for conforming approximations of interface problems has C_rel·C_eff bounded independently of the contrast M (≳ M^{1/4} at a checkerboard cross-point). Abstract read only.
- **Status in the literature.** I found no conforming equilibrated estimator proved jump-robust without quasi-monotonicity; the Cai–Cai–Zhang conforming results assume it. That is an abstract-level check.
- **Numerics: jumps.** With an aligned jump between two subdomains (a quasi-monotone case), effectivities do not depend on the contrast up to 10⁴. Checkerboard (non-quasi-monotone) data are not tested.
- **Numerics: within-cell variation.** ε_lo,K costs a factor (ε_hi,K/ε_lo,K)^{1/2} in the constants. For smooth ε this tends to 1 under refinement.

Literature and novelty.

- **Variable reaction.** Ainsworth–Vejchodský [AV14] treat a piecewise-constant reaction coefficient (any dimension, mixed boundary conditions), robustly in the size of the reaction. That is a robustness in the reaction, not in the diffusion contrast. (Scope corrected, audit R2.)
- **Variable diffusion.** ESV10 allow inhomogeneous diffusion in DG, with constants depending on the local contrast.
- **Smoothed fluxes.** Using the L^2 projection $\Pi_1\varepsilon$ in the equilibrated flux, so that the remainder is $O(h^2)$ while the compatibility at P1 tests is kept, is the natural choice. I did not find it stated for algebraic stabilisations.
- **Verdict:** an adaptation, not a new result.

Numerics (2D smoke runs, BJK, uniform meshes n = 8, 16; c = 0 and 1). The table compares the SV estimator with the gap 2 bound `robust` (columns marked "gap 2"); each cell gives the range over the runs.

ε(x)	ε	contrast	η_SV/|||e|||	κ SV/|||e|||_⊕	SV_E/E	gap 2: η_E/E	η_ε/η
smooth, ε·k^{x₁}	10^{-2}	1	1.7–4.6	3.5–4.1	3.0–3.1	3.1–3.9	0
smooth	10^{-2}	10^2	1.1–1.2	3.2–3.4	3.2–3.3	7.1–7.2	0.020–0.036
smooth	10^{-2}	10^4	1.2–1.4	3.5–4.3	3.5–4.3	10.9–12.8	0.071–0.117
smooth	10^{-6}	10^2–10^4	—	4.2–5.5	3.8–4.5	3.8–6.1	≤ 0.006
jump at $x_1 = ½$	10^{-2}	10^2, 10^4	4.0–8.1	3.8–4.5	2.8–3.4	2.9–4.0	0
jump	10^{-6}	10^2, 10^4	—	3.4–5.1	3.4–5.0	3.5–5.1	0
3D smooth (4913 vertices)	10^{-2}	10^2	1.1	3.1	3.0–3.1	6.9	—
3D jump (4913 vertices)	10^{-2}	10^2	3.6–4.1	4.2–4.3	3.2	3.3–4.0	0

- **Guarantee.** All bounds held in every run, in 2D and 3D, and in SV-marked adaptive loops with adaptive stopping (contrast 10^3).
- **Rate of η_ε.** η_ε/η halves from n = 8 to n = 16: it is $O(h^2)$, against an error of $O(h)$.
- **The P1 projection matters.** With the cell mean $\bar{\varepsilon}_K$ in the flux (the first version tried), smooth contrast 10^4 gave κ SV/|||e|||_⊕ = 8.3–8.9 and η_ε/η ≈ 0.4.
- **The ABR17-based bounds** (ESV, ABR17, gap 2 `robust`) use the cruder cell-mean version. Their E-effectivity grows to 11–13 at contrast 10^4; the SV estimator is the one to use for variable ε.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_variable_eps` and `::test_eps_cell_bounds`. The problem is `get_problem("variable_eps", kind="smooth" | "jump", contrast=…)`.

References added.

- [BV00] C. Bernardi, R. Verfürth, Adaptive finite element methods for elliptic equations with non-smooth coefficients, Numer. Math. 85 (2000) 579–608. [from memory — verify]
- [Pe02] M. Petzoldt, A posteriori error estimators for elliptic equations with discontinuous coefficients, Adv. Comput. Math. 16 (2002) 47–75. [from memory — verify]
- Cai, He, Zhang and Cai, Cai, Zhang: robust estimators for interface problems (arXiv 1603.01024, arXiv 2001.09102). [abstracts]
- [AV14] see §8.